HM62256 Series — Maintenance only

32768-word × 8-bit High Speed CMOS Static RAM

Features

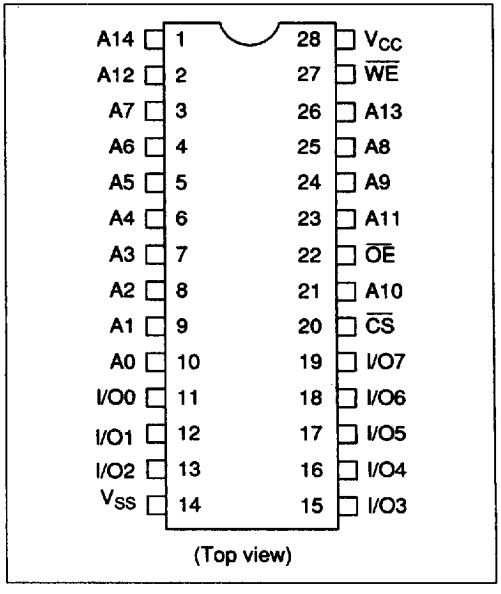
- High speed: Fast access time 85/100/120/150 ns (max)
- Low power standby and low power operation
  - — Standby: 200 µW (typ)/ 10 µW (typ) (L-/L-SL-version)
  - — Operation: 40 mW (typ) (f = 1 MHz)
- Single 5 V supply
- Completely static RAM: No clock or timing strobe required
- Equal access and cycle time
- Common data input and output, three-state output
- Directly TTL compatible—all inputs and outputs
- Battery back up operation capability (L-/L-SL-version)

Ordering Information

Type No.	Access time	Package
HM62256P-8	85 ns	600-mil 28-pin plastic DIP (DP-28)
HM62256P-10	100 ns	
HM62256P-12	120 ns	
HM62256P-15	150 ns	
HM62256LP-8	85 ns	
HM62256LP-10	100 ns	
HM62256LP-12	120 ns	
HM62256LP-15	150 ns	
HM62256LP-10SL	100 ns	
HM62256LP-12SL	120 ns	
HM62256LP-15SL	150 ns	

Type No.	Access time	Package
HM62256FP-8T	85 ns	28-pin plastic SOP (FP-28DA)
HM62256FP-10T	100 ns	
HM62256FP-12T	120 ns	
HM62256FP-15T	150 ns	
HM62256LFP-8T	85 ns	
HM62256LFP-10T	100 ns	
HM62256LFP-12T	120 ns	
HM62256LFP-15T	150 ns	
HM62256LFP-10SLT	100 ns	
HM62256LFP-12SLT	120 ns	
HM62256LFP-15SLT	150 ns	

Pin Arrangement



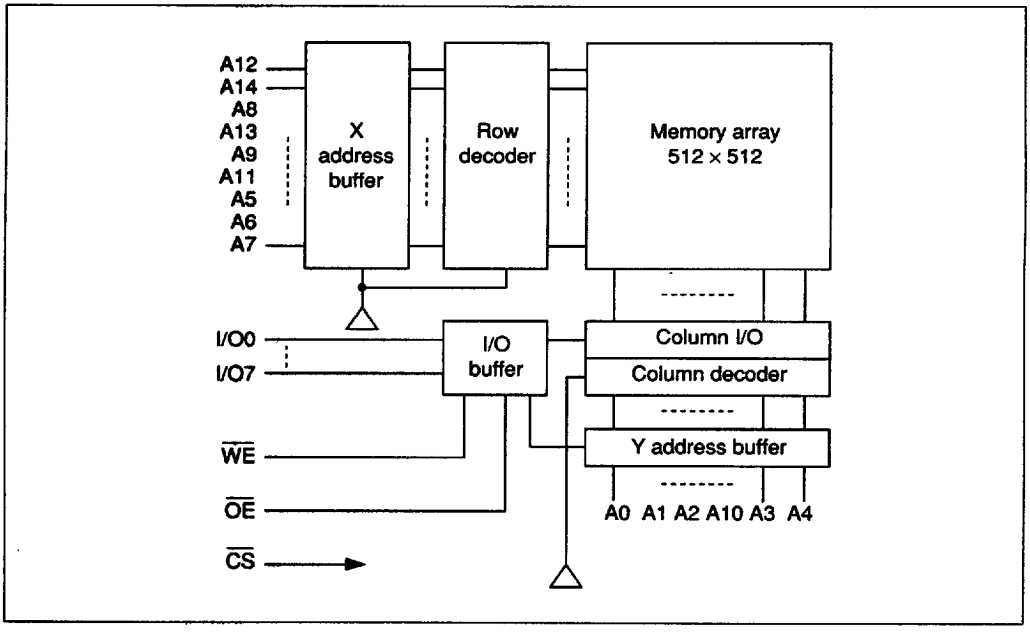
(Top view)

Note: This device is not available for new application.

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Block Diagram



Truth Table

$\overline{CS}$	$\overline{OE}$	$\overline{WE}$	Mode	V_{CC} current	I/O pin	Reference cycle
H	x	x	Not selected	I_{SB}, I_{SB1}	High Z	—
L	L	H	Read	I_{CC}	Dout	Read cycle No. 1–3
L	H	L	Write	I_{CC}	Din	Write cycle No. 1
L	L	L	Write	I_{CC}	Din	Write cycle No. 2

Note: x means H or L

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Absolute Maximum Ratings

Parameter	Symbol	Rating	Unit
Voltage on any pin relative to V_{SS}	V_T	−0.5* to +7.0	V
Power dissipation	P_T	1.0	W
Operating temperature	T_{opr}	0 to +70	°C
Storage temperature	T_{stg}	−55 to +125	°C
Temperature under bias	T_{bias}	−10 to +85	°C

Note: −3.0 V min for pulse width ≤ 50 ns

Recommended DC Operating Conditions (Ta = 0 to +70°C)

Parameter	Symbol	Min	Typ	Max	Unit
Supply voltage	V_{CC}	4.5	5.0	5.5	V
	V_{SS}	0	0	0	V
Input voltage	V_{IH}	2.2	—	6.0	V
	V_{IL}	−0.5*	—	0.8	V

Note: −3.0 V min for pulse width ≤ 50 ns

DC Characteristics (V_{CC} = 5 V ± 10%, V_{SS} = 0 V, Ta = 0 to +70°C)

Parameter		Symbol	Min	Typ *1	Max	Unit	Test condition
Input leakage current		$\|I_{LI}\|$	—	—	2	µA	$V_{IN} = V_{SS}$ to V_{CC}
Output leakage current		$\|I_{LO}\|$	—	—	2	µA	$\overline{CS} = V_{IH}$ or $\overline{OE} = V_{IH}$ or $\overline{WE} = V_{IL}$ $V_{I/O} = V_{SS}$ to V_{CC}
Operating power supply current		I_{CC}	—	8	15	mA	$\overline{CS} = V_{IL}$, $I_{I/O}$ = 0 mA
Average operating power supply current	HM62256-8	I_{CC1}	—	50	70	mA	Min. cycle, duty = 100%, $\overline{CS} = V_{IL}$, $I_{I/O}$ = 0 mA
	HM62256-10		—	40	70	mA	
	HM62256-12		—	35	70	mA	
	HM62256-15		—	33	70	mA	
		I_{CC2}	—	8	15	mA	$\overline{CS} = V_{IL}$, $V_{IH} = V_{CC}$, V_{IL} = 0V, $I_{I/O}$ = 0 mA f = 1 MHz

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DC Characteristics (V_{CC} = 5 V ± 10%, V_{SS} = 0 V, Ta = 0 to +70°C) (cont)

Parameter	Symbol	Min	Typ *1	Max	Unit	Test condition
Standby power supply current	I_{SB}	—	0.5	3	mA	$\overline{CS}$ = V_{IH}
	I_{SB1}	—	0.04	2	mA	$\overline{CS} \geq V_{CC} - 0.2V$, $0V \leq V_{IN}$
		—	2*2	100*2	µA	
		—	2*3	50*3		
Output voltage	V_{OL}	—	—	0.4	V	I_{OL} = 2.1 mA
	V_{OH}	2.4	—	—	V	I_{OH} = −1.0 mA

Notes: 1. Typical values are at V_{CC} = 5.0 V, Ta = 25°C and specified loading.
2. These characterisitcs are guaranteed only for L-version.
3. These characterisitcs are guaranteed only for L-SL version.

Capacitance (Ta = 25°C, f= 1 MHz)

Parameter	Symbol	Typ	Max	Unit	Test Condition
Input capacitance	C_{IN}	—	6	pF	V_{IN} = 0 V
Input/output capacitance	$C_{I/O}$	—	8	pF	$V_{I/O}$ = 0 V

Note: These parameters are sampled and not 100% tested.

AC Characteristics (V_{CC} = 5 V ± 10%, Ta = 0 to +70°C unless otherwise noted)

AC Test Conditions:

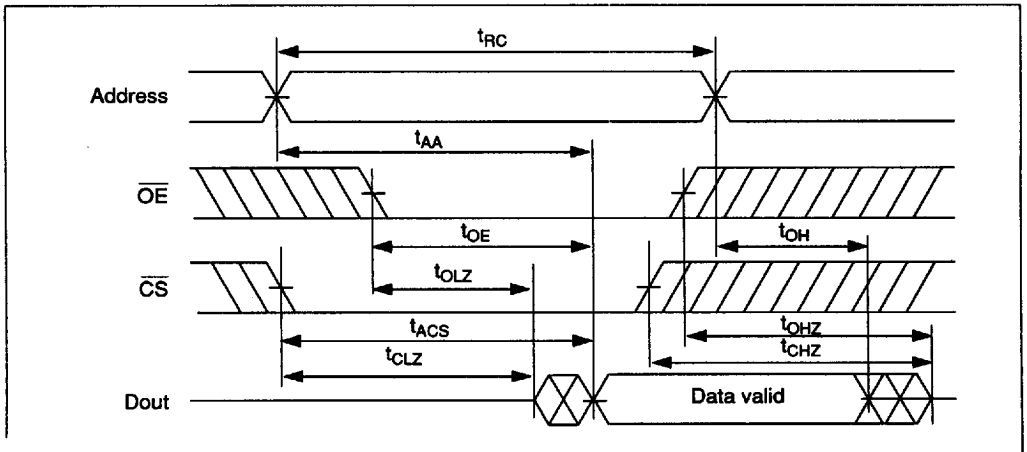
- Input pulse levels: 0.8 V to 2.4 V
- Input and output timing reference levels: 1.5 V
- Input rise and fall times: 5 ns
- Output load: 1TTL gate and C_L = 100 pF (including scope and jig)

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Read Cycle

		HM62256-8		HM62256-10		HM62256-12		HM62256-15		
Parameter	**Symbol**	**Min**	**Max**	**Min**	**Max**	**Min**	**Max**	**Min**	**Max**	**Unit**
Read cycle time	t_{RC}	85	—	100	—	120	—	150	—	ns
Address access time	t_{AA}	—	85	—	100	—	120	—	150	ns
Chip select access time	t_{ACS}	—	85	—	100	—	120	—	150	ns
Output enable to output valid	t_{OE}	—	45	—	50	—	60	—	70	ns
Output hold from address change	t_{OH}	5	—	10	—	10	—	10	—	ns
Chip selection to output in low Z	t_{CLZ}	10	—	10	—	10	—	10	—	ns
Output enable to output in low Z	t_{OLZ}	5	—	5	—	5	—	5	—	ns
Chip deselection to output in high Z	t_{CHZ}	0	30	0	35	0	40	0	50	ns
Output disable to output in high Z	t_{OHZ}	0	30	0	35	0	40	0	50	ns

Read Timing Waveform (1)

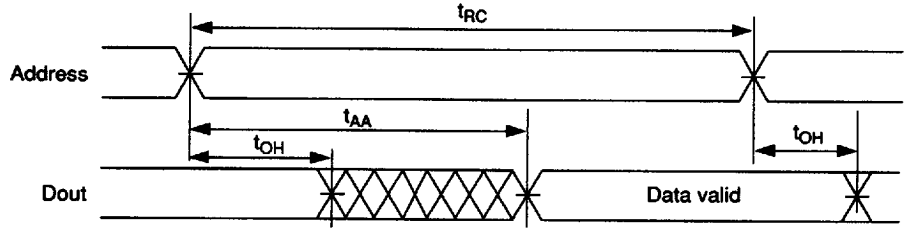


Note: $\overline{WE}$ is high during the read cycle.

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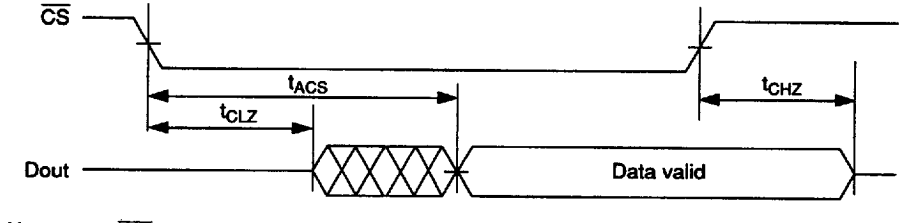
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Read Timing Waveform (2)



Notes: 1. $\overline{WE}$ is high during the read cycle.
2. Device is continuously selected, $\overline{CS}$ = V_{IL}.
3. $\overline{OE}$ = V_{IL}.

Read Timing Waveform (3)



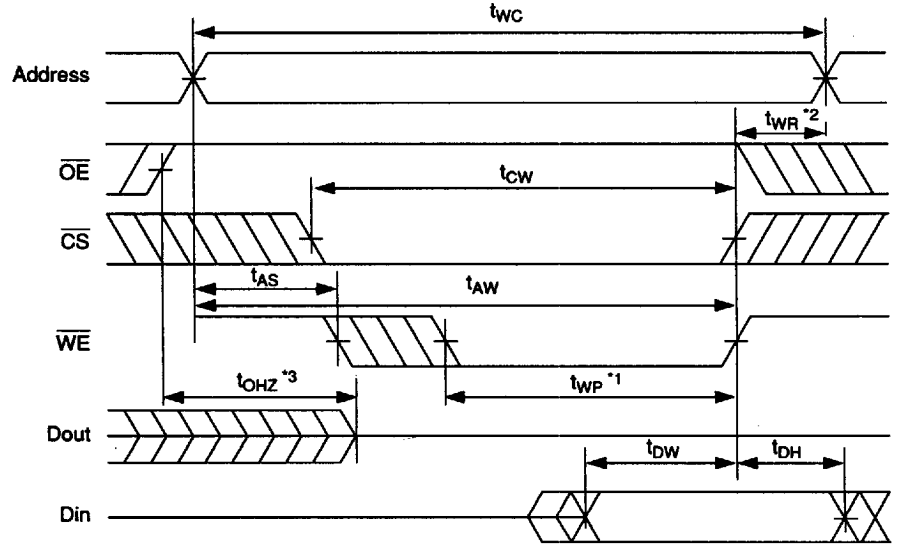
Notes: 1. $\overline{WE}$ is high during the read cycle.
2. Address valid prior to or coincident with $\overline{CS}$ transition low.
3. $\overline{OE}$ = V_{IL}.

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Write Cycle

Parameter	Symbol	HM62256-8 Min	HM62256-8 Max	HM62256-10 Min	HM62256-10 Max	HM62256-12 Min	HM62256-12 Max	HM62256-15 Min	HM62256-15 Max	Unit
Write cycle time	t_{WC}	85	—	100	—	120	—	150	—	ns
Chip selection to end of write	t_{CW}	75	—	80	—	85	—	100	—	ns
Address valid to end of write	t_{AW}	75	—	80	—	85	—	100	—	ns
Address set up time	t_{AS}	0	—	0	—	0	—	0	—	ns
Write pulse width	t_{WP}	60	—	60	—	70	—	90	—	ns
Write recovery time	t_{WR}	10	—	0	—	0	—	0	—	ns
Write to output in high Z	t_{WHZ}	0	30	0	35	0	40	0	50	ns
Data to write time overlap	t_{DW}	40	—	40	—	50	—	60	—	ns
Data hold from write time	t_{DH}	0	—	0	—	0	—	0	—	ns
Output disable to output in high Z	t_{OHZ}	0	30	0	35	0	40	0	50	ns
Output active from end of write	t_{OW}	5	—	5	—	5	—	5	—	ns

Write Timing Waveform (1) ($\overline{OE}$ Clock)

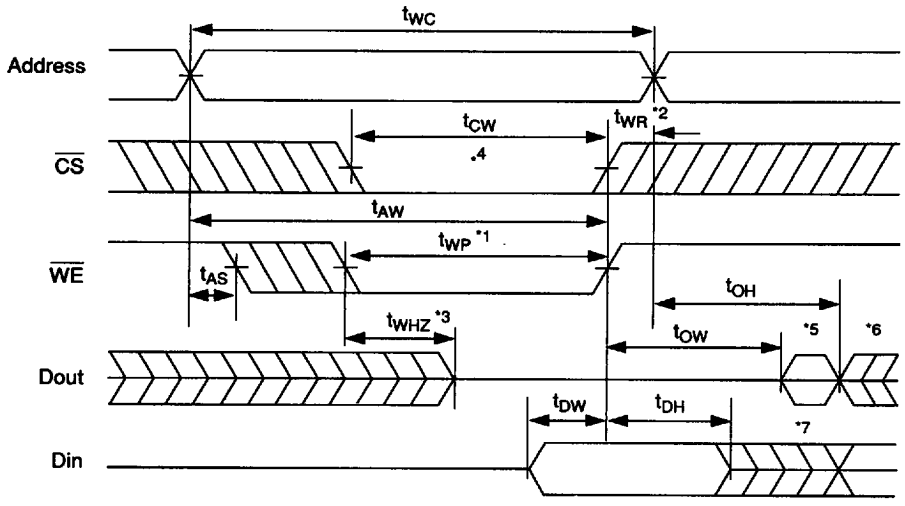


Notes: 1. A write occurs during the overlap (t_{WP}) of a low $\overline{CS}$ and a low $\overline{WE}$.
2. t_{WR} is measured from the earlier of $\overline{CS}$ or $\overline{WE}$ going high to the end of write cycle.
3. During this period, I/O pins are in the output state. Out of phase input signals must not be applied.

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Write Timing Waveform (2) ($\overline{OE}$ Fixed Low)



Notes: 1. A write occurs during the overlap (t_{WP}) of a low $\overline{CS}$ and a low $\overline{WE}$.
2. t_{WR} is measured from the earlier of $\overline{CS}$ or $\overline{WE}$ going high to the end of write cycle.
3. During this period, I/O pins are in the output state. The input signals out of phase must not be applied.
4. If the $\overline{CS}$ low transistion occurs simultaneously with the $\overline{WE}$ low transition or after the $\overline{WE}$ low transition, outputs remain in a high impedance state.
5. Dout is in the same phase of written data of this write cycle.
6. Dout is the read data of next address.
7. If $\overline{CS}$ is low during this period, I/O pins are in the output state. Out of phase input signals must not be appplied to I/O pins.

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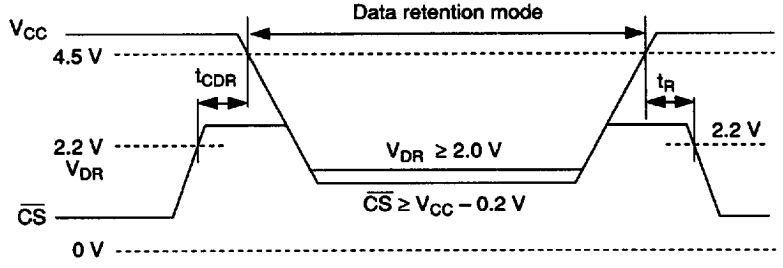
Low V_{CC} Data Retention Characteristics (Ta = 0 to +70°C)

These characteristics are guaranteed only for L- and L-SL version.

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
V_{CC} for data retention	V_{DR}	$\overline{CS} \geq V_{CC} - 0.2$ V	2.0	—	—	V
Data retention current	I_{CCDR}	V_{CC} = 3.0 V, $\overline{CS} \geq 2.8$ V, 0 V ≤ V_{IN}	—	—	50 *2	µA
			—	—	10 *3	
Chip deselect to data retention time	t_{CDR}	See retention waveform	0	—	—	ns
Operation recovery time	t_R	See retention waveform	t_{RC} *1	—	—	ns

Notes: 1. t_{RC} = read cycle time
2. These characteristics are guaranteed only for L-version, V_{IL} = –0.3 V min, 20 µA max. at Ta = 0 to 40°C.
3. These characteristics are guaranteed only for L-SL version, V_{IL} = –0.3 V min, 3 µA max. at Ta = 0 to 40°C.

Low V_{CC} Data Retention Waveform

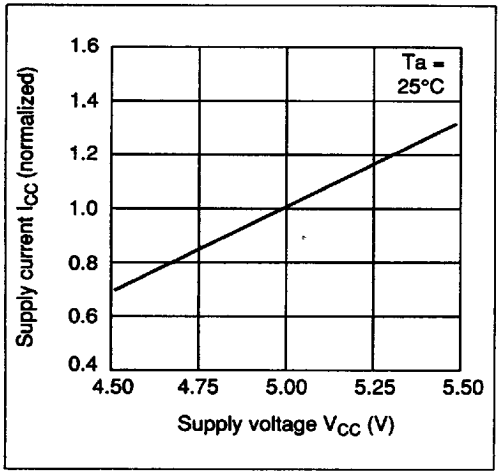


Note: In data retention mode, $\overline{CS}$ controls the address, $\overline{WE}$, $\overline{OE}$, and Din buffers. V_{in} for these inputs can be in high impedance state in data retention mode.

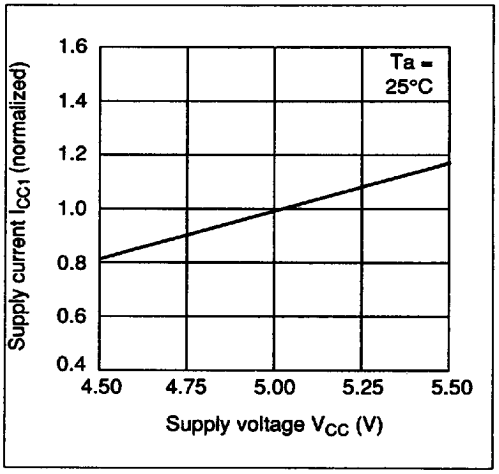
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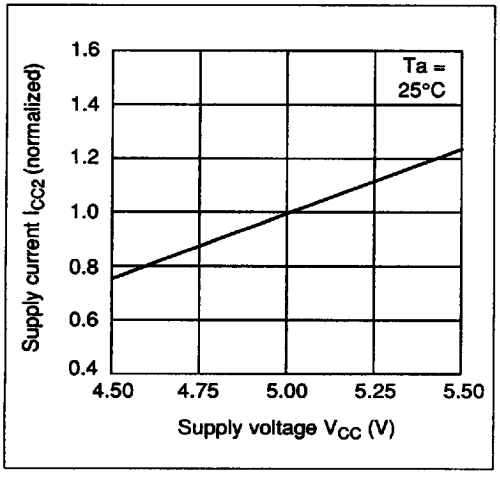
Characteristic Curves



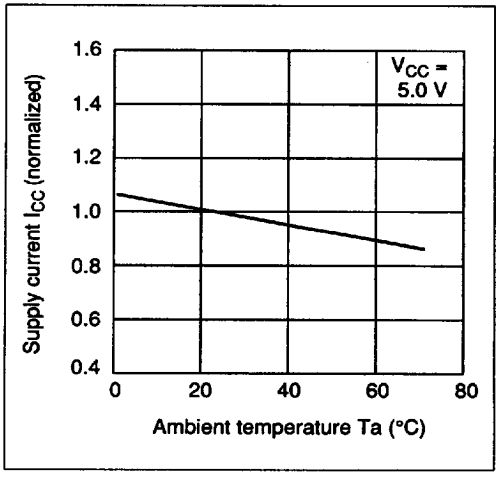
Supply Current vs. Supply Voltage (1)



Supply Current vs. Supply Voltage (2)

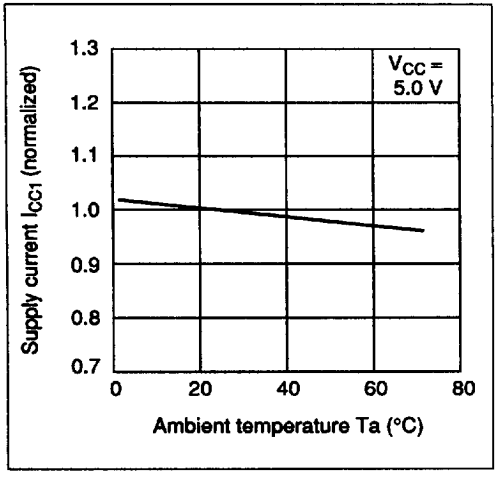


Supply Current vs. Supply Voltage (3)

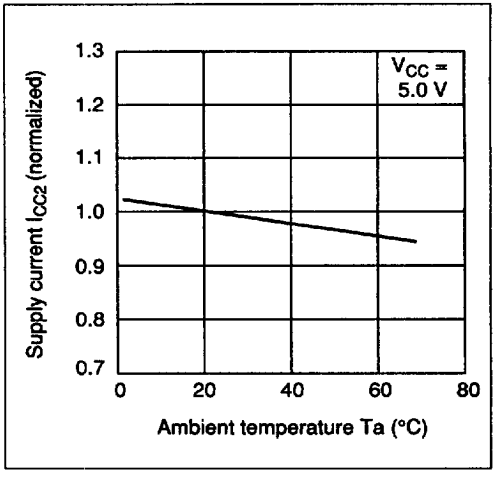


Supply Current vs. Ambient Temperature (1)

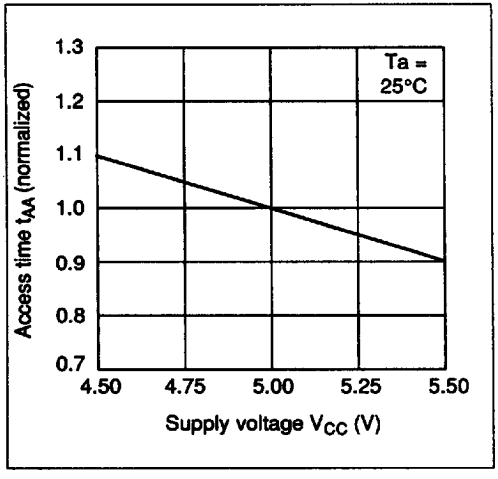
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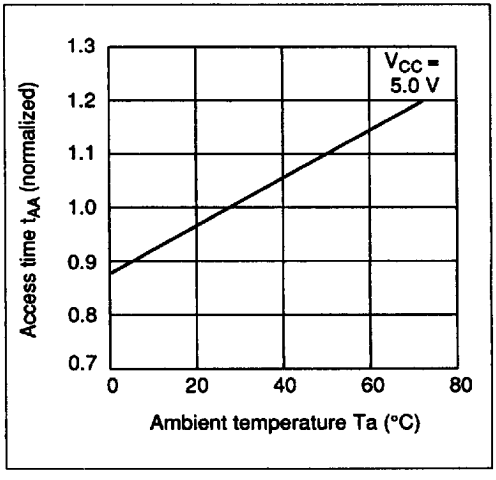
Supply Current vs. Ambient Temperature (2)



Supply Current vs. Ambient Temperature (3)



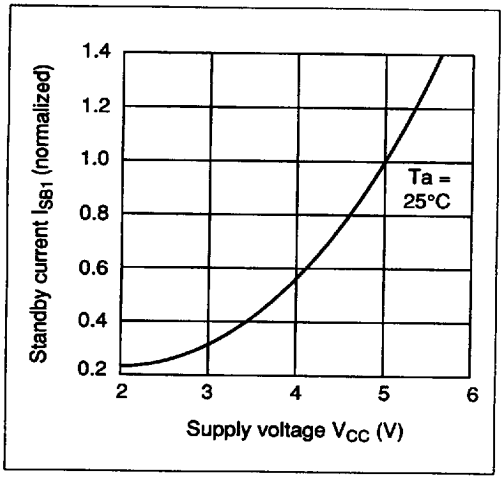
Access Time vs. Supply Voltage



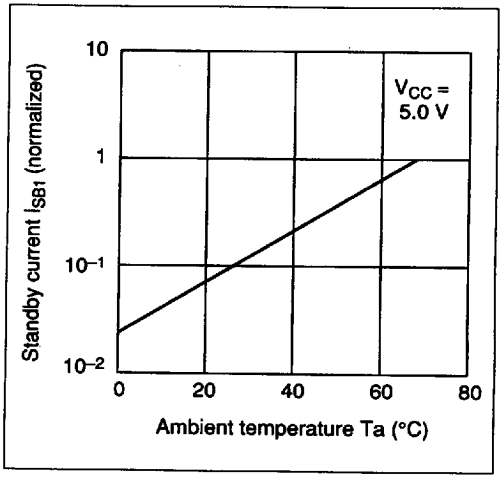
Access Time vs. Ambient Temperature

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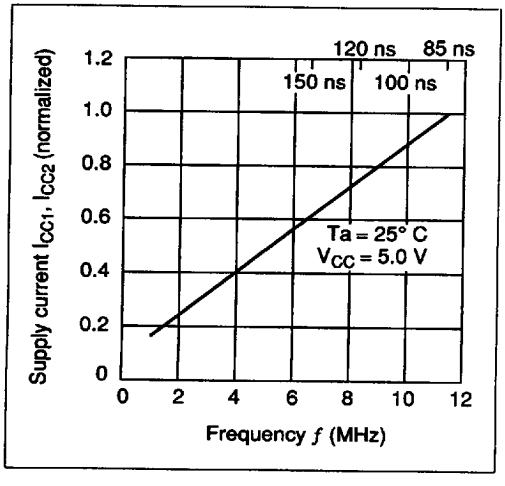
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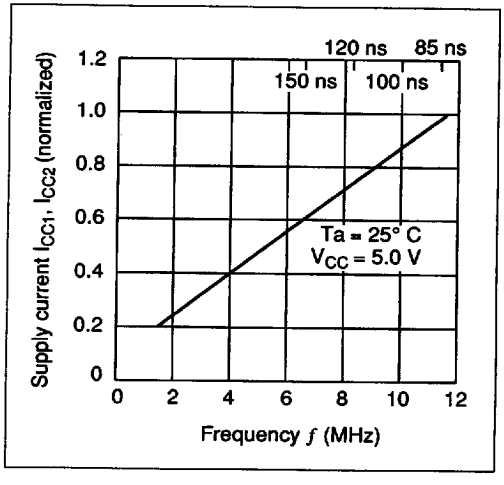
Standby Current vs. Supply Voltage



Standby Current vs. Ambient Temperature

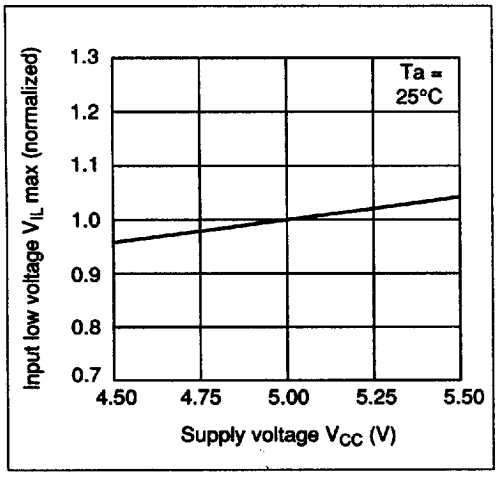


Supply Current vs. Frequency (Read)

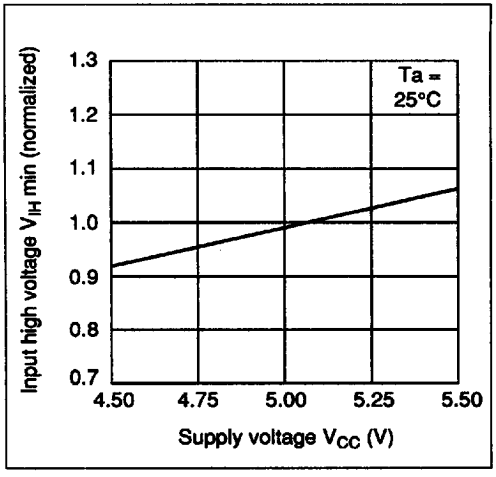


Supply Current vs. Frequency (Write)

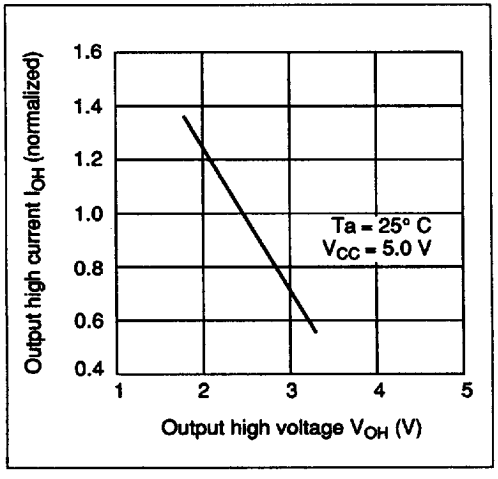
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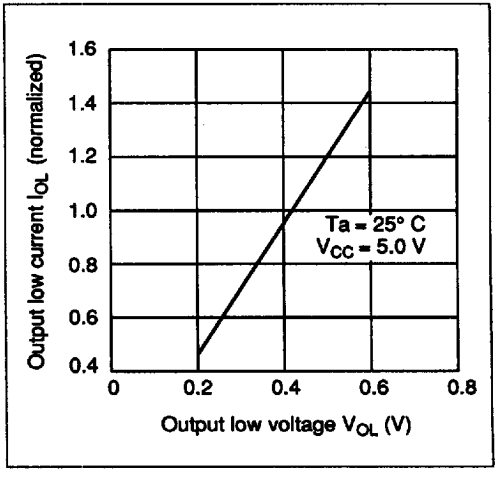
Input Low Voltage vs. Supply Voltage



Input High Voltage vs. Supply Voltage



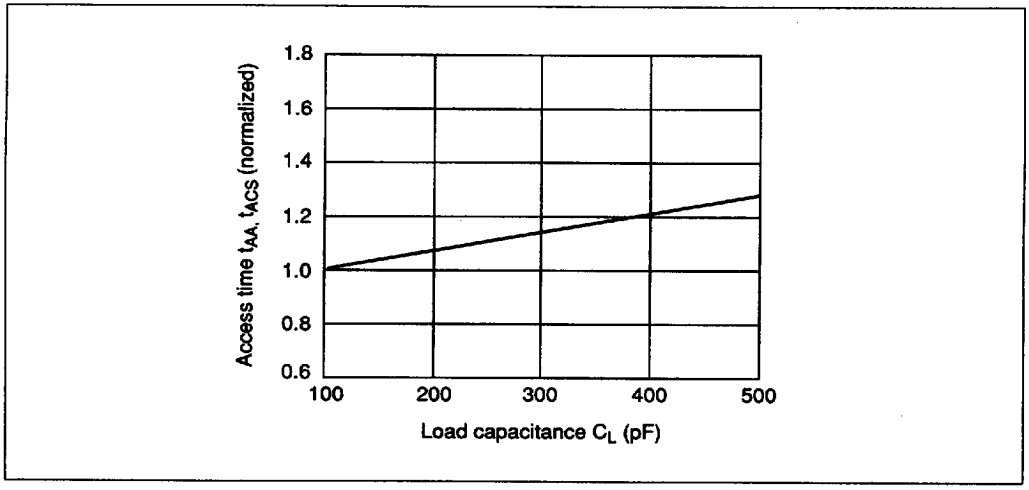
Output Current vs. Output Voltage (High)



Output Current vs. Output Voltage (Low)

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Access Time vs. Load Capacitance

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